



BUNDI

AI-Driven Bio-Acoustic Surveillance

Digitizing Conservation in the Kakamega Forest

Team Bundi | Environmental Conservation & Smart IoT

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| Team & Mission

Who We Are

We are a dedicated team of Mechatronic Engineers and Data Scientists. We specialize in embedded AI and IoT solutions designed specifically for complex environmental challenges.

Our Mission

To transition forestry protection from reactive damage assessment to proactive real-time interception. We are deploying "smart ears" into the forest to stop illegal activities before they cause irreversible damage.

| The Crisis in Kakamega Forest



The "Ghost" Problem

Illegal logging and charcoal burning happen deep in the forest, often invisible to satellite imagery until weeks later when the damage is done.



Resource Scarcity

KFS Rangers are vastly outnumbered and forced to patrol randomly, leading to high fuel costs and critically low interception rates.



The "Trust Gap"

Local communities witness crimes but fear reporting them due to a lack of anonymity and eroding trust in the reporting systems.

The Solution: BUNDI

The Forest's Digital Nervous System

A network of solar-powered acoustic sensors creating a 24/7 "listening grid".

- **Detect:** Edge AI identifies chainsaws & vehicles in < 60s.
- **Connect:** LoRa Mesh network transmits alerts (No SIMs).
- **Verify:** Rangers get GPS coordinates on a dashboard.
- **Empower:** Anonymous community reporting via USSD/Blockchain.



Technical Architecture

The Edge (Hardware)

- **Brain:** ESP32-S3 (TensorFlow Lite for Microcontrollers)
- **Ears:** INMP441 / ICS-43434 High-Fidelity MEMS Mics
- **Voice:** SX1278 LoRa Radio (5-15km range)
- **Power:** Solar + Deep Sleep (Wake-on-Sound)

The Cloud (Software)

- **AI Model:** CRNN + XGBoost (Forest dataset trained)
- **Dashboard:** Flask-based "Commander" interface
- **Trust Layer:** Blockchain Ledger for immutable reporting

| Prototype & Validation (TRL 6)

90%

Detection Accuracy

For mechanical sounds like chainsaws and motorbikes in pilot tests.

<60s

Alert Latency

Reduced from days (satellite) to mere seconds (acoustic).

Status: V2.1 Command Center deployed with 15 Strategic Nodes mapped.

| Impact & Scalability Model



Impact: Saving just 5 mature Elgon Teak trees covers the hardware cost for 50 acres of protection.

The Vision

To make BUNDI the standard for "Smart Forestry" in East Africa—empowering rangers, engaging communities, and protecting our carbon sinks.

The Ask

Partnership with KFS/KWS & Seed Funding for 50 Units

| Image Sources



<https://envirotecmagazine.com/wp-content/uploads/2022/09/Funding-for-early-detection.jpg>

Source: envirotecmagazine.com



https://images.stockcake.com/public/6/b/4/6b4246eb-93b9-4b9b-893e-7e89ca9a8f61_large/digital-forest-network-stockcake.jpg

Source: stockcake.com



<https://lp-cms-production.imgix.net/2023-06/shutterstock1469073227.jpg?w=1920&h=640&fit=crop&crop=faces%2Cedges&auto=format&q=75>

Source: www.lonelyplanet.com